

Homework 1 - Foundations of proof

Practice submission A | Student: Fictional learner

Question 1. Sets in three forms

(a) $A = \{16, 25, 27, 36, 49, 64, 81\}$.

(b) $B = \{5k + 1 : k \in \mathbb{Z}, k \geq 0\}$.

(c) $C = \{3^k : k \in \mathbb{Z}\}$.

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Question 2. A divisibility claim

False: take $a=2$ and $b=5$. Their product is $10=10 \cdot 1$. Neither $2/10$ nor $5/10$ is an integer, so neither factor is divisible by 10. This disproves the universal implication.

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Question 3. A rational point inside an interval

My first thought was to call this a weighted average, but I will check the required definition directly. Choose integers a, b, c, d with $b, d \neq 0$ such that $x = a/b$ and $y = c/d$.

Then $u = (3ad + bc)/(4bd)$. Both parts are integers, and $4bd$ is nonzero. Hence $u \in \mathbb{Q}$.

Finally, $u - x = (y - x)/4 > 0$ and $y - u = 3(y - x)/4 > 0$, since $x < y$. This proves the strict inequalities.

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Question 4. An odd expression

For any integer n , first factor $n^2+3n=n(n+3)$. If n is even, this product is even. If n is odd, write $n=2k+1$ with $k \in \mathbb{Z}$; then $n+3=2(k+2)$, so the product is again even.

Thus $n(n+3)=2m$ for some integer m , and $n^2+3n+1=2m+1$. This is odd by definition. No positivity assumption was made.

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Question 5. Removing either set

Let $t \in A \setminus (B \cup C)$. It lies in A but in neither B nor C , so it lies in both differences on the right. This proves left-to-right containment.

Now let t belong to the intersection on the right. Membership in its two factors says $t \in A$, $t \notin B$, and $t \notin C$. Thus it lies in A but outside $B \cup C$, which is the left-hand side. Both containments hold, so the sets are equal.